

PRAKTIKUM BASIS DATA

MODUL 13

NoSQL Document Database



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PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK

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UNIVERSITAS TELKOM

2026

Jurnal

1. Tambahkan 1 toko baru bebas ke koleksi toko, dengan ketentuan:
 - Minimal memiliki 2 produk di dalam array products
 - Isi semua field: name, category, city, address, products

Jawaban:

Query

```
db.toko.insertOne({
  "name" : "Toko Sinar Jaya",
  "category" : "Fashion",
  "city" : "Surabaya",
  "address" : { "street" : "Jl. Pemuda", "zipcode" : "60271" },
  "products" : [
    { "item" : "Kaos Polos", "price" : 75000, "stock" : 50 },
    { "item" : "Jaket Hoodie", "price" : 250000, "stock" : 15 }
  ]
})
```

Screenshot

```
> db.toko.insertOne({
  "name" : "Toko Sinar Jaya",
  "category" : "Fashion",
  "city" : "Surabaya",
  "address" : { "street" : "Jl. Pemuda", "zipcode" : "60271" },
  "products" : [
    { "item" : "Kaos Polos", "price" : 75000, "stock" : 50 },
    { "item" : "Jaket Hoodie", "price" : 250000, "stock" : 15 }
  ]
})
< {
  acknowledged: true,
  insertedId: ObjectId('6a12f3d77aae4b29fd4724e5')
}
```

2. Tampilkan semua toko yang berada di kota Jakarta.

Jawaban:

Query

```
db.toko.find({ "city": "Jakarta" }).pretty()
```

Screenshot



```
> db.toko.find({ "city": "Jakarta" }).pretty()
< {
  _id: ObjectId('6a12f37e7aae4b29fd4724e1'),
  name: 'Toko Berkah',
  category: 'Elektronik',
  city: 'Jakarta',
  address: {
    street: 'Jl. Sudirman',
    zipcode: '10220'
  },
  products: [
    {
      item: 'TV',
      price: 3500000,
      stock: 10
    },
    {
      item: 'Kulkas',
      price: 2800000,
      stock: 5
    }
  ]
}

{
  _id: ObjectId('6a12f37e7aae4b29fd4724e3'),
  name: 'Toko Salim Bakery',
  category: 'Elektronik',
  city: 'Jakarta',
  address: {
    street: 'Jl. Sudirman',
    zipcode: '10220'
  },
  products: [
    {
      item: 'Monitor LED',
      price: 7700000,
      stock: 11
    },
    {
      item: 'Airfryer Panasonic',
      price: 4850000,
      stock: 7
    }
  ]
}
test>
```

3. Tampilkan toko yang memiliki produk dengan price lebih dari 3.000.000.

Jawaban:

Query

```
db.toko.find({
  "products.price": { $gt: 3000000 }
}).pretty()
```

Screenshot



4. Ubah category Toko Berkah dari Fashion menjadi Aksesoris. Tambahkan 1 produk baru ke dalam array products milik Toko Maju menggunakan \$push.

Jawaban:

Query

```
db.toko.updateOne(
  { "name": "Toko Berkah" },
  { $set: { "category": "Aksesoris" } }
)
```

```
db.toko.updateOne(
  { "name": "Toko Maju" },
  {
    $push: {
      "products": {
        "item": "Mouse Wireless",
        "price": 250000,
        "stock": 30
      }
    }
  }
)
```

Screenshot



The image shows two side-by-side screenshots of a MongoDB command-line interface. The left screenshot shows a query to update the 'category' of 'Toko Berkah' and its result. The right screenshot shows a query to push a new product into the 'products' array of 'Toko Maju' and its result.

```
> db.toko.updateOne(
  { "name": "Toko Berkah" },
  { $set: { "category": "Aksesoris" } }
)
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
test>
```

```
> db.toko.updateOne(
  { "name": "Toko Maju" },
  {
    $push: {
      "products": {
        "item": "Mouse Wireless",
        "price": 250000,
        "stock": 30
      }
    }
  }
)
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
test>
```

5. Hitung jumlah toko per city, urutkan dari kota yang paling banyak tokonya.

Jawaban:

Query

```

db.toko.aggregate([
  {
    $group: {
      _id: "$city",
      jumlah_toko: { $sum: 1 }
    }
  },
  {
    $sort: {
      jumlah_toko: -1
    }
  }
])

```

Screenshot

```

> db.toko.aggregate([
  {
    $group: {
      _id: "$city",
      jumlah_toko: { $sum: 1 }
    }
  },
  {
    $sort: {
      jumlah_toko: -1
    }
  }
])
< {
  _id: 'Surabaya',
  jumlah_toko: 2
}
{
  _id: 'Jakarta',
  jumlah_toko: 2
}
{
  _id: 'Bandung',
  jumlah_toko: 1
}
test>

```

6. Hitung rata-rata harga produk per category.

Jawaban:

Query

```

db.toko.aggregate([
  {
    $unwind: "$products"
  },
  {
    $group: {
      _id: "$category",
      rata_rata_harga: { $avg: "$products.price" }
    }
  },
  {
    $sort: {
      rata_rata_harga: -1
    }
  }
])

```

Screenshot

```

> db.toko.aggregate([
  {
    $unwind: "$products"
  },
  {
    $group: {
      _id: "$category",
      rata_rata_harga: { $avg: "$products.price" }
    }
  },
  {
    $sort: {
      rata_rata_harga: -1
    }
  }
])
< {
  _id: 'Elektronik',
  rata_rata_harga: 6114285.714285715
}
{
  _id: 'Aksesoris',
  rata_rata_harga: 3150000
}
{
  _id: 'Fashion',
  rata_rata_harga: 162500
}
test>

```

7. Tambahkan 1 produk baru ke dalam array products milik Toko Maju menggunakan \$push.

Jawaban:

Query

```

db.toko.updateOne(
  { "name": "Toko Maju" },
  {
    $push: {
      "products": {
        "item": "Keyboard Mechanical",
        "price": 650000,
        "stock": 12
      }
    }
  }
)

```

Screenshot

```

> db.toko.updateOne(
  { "name": "Toko Maju" },
  {
    $push: {
      "products": {
        "item": "Keyboard Mechanical",
        "price": 650000,
        "stock": 12
      }
    }
  }
)
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
test>

```

8. Tampilkan harga produk tertinggi (\$max) per city.

Jawaban:

Query


```

db.toko.aggregate([
  {
    $unwind: "$products"
  },
  {
    $group: {
      _id: "$city",
      harga_produk_tertinggi: { $max: "$products.price" }
    }
  },
  {
    $sort: {
      harga_produk_tertinggi: -1
    }
  }
])

```

Screenshot

```

> db.toko.aggregate([
  {
    $unwind: "$products"
  },
  {
    $group: {
      _id: "$city",
      harga_produk_tertinggi: { $max: "$products.price" }
    }
  },
  {
    $sort: {
      harga_produk_tertinggi: -1
    }
  }
])
< {
  _id: 'Surabaya',
  harga_produk_tertinggi: 12000000
}
{
  _id: 'Bandung',
  harga_produk_tertinggi: 8000000
}
{
  _id: 'Jakarta',
  harga_produk_tertinggi: 7700000
}
test>

```

9. Buat pipeline lengkap dengan 4 tahap:

- \$match — filter hanya toko category Elektronik
- \$unwind — bongkar array products
- \$group — hitung total stock per city
- \$sort — urutkan dari total stock terbesar

Jawaban:

Query

```
db.toko.aggregate([
  {
    $match: {
      "category": "Elektronik"
    }
  },
  {
    $unwind: "$products"
  },
  {
    $group: {
      _id: "$city",
      total_stock: { $sum: "$products.stock" }
    }
  },
  {
    $sort: {
      total_stock: -1
    }
  }
])
```

Screenshot

```
> db.toko.aggregate([
  {
    $match: {
      "category": "Elektronik"
    }
  },
  {
    $unwind: "$products"
  },
  {
    $group: {
      _id: "$city",
      total_stock: { $sum: "$products.stock" }
    }
  },
  {
    $sort: {
      total_stock: -1
    }
  }
])
< {
  _id: 'Bandung',
  total_stock: 65
}
{
  _id: 'Surabaya',
  total_stock: 29
}
{
  _id: 'Jakarta',
  total_stock: 18
}
test>
```

10. Hapus toko yang memenuhi dua kondisi sekaligus:

- category adalah Fashion
- city adalah Surabaya

Jawaban:

Query

```
db.toko.deleteMany({
  "category": "Fashion",
  "city": "Surabaya"
})
```

Screenshot

```
> db.toko.deleteMany({
  "category": "Fashion",
  "city": "Surabaya"
})
< {
  acknowledged: true,
  deletedCount: 1
}
test> |
```